Automated Testing

@Cognite for Marathon

Two strategies for testing project deliveries

- Testing DATA
- Testing the full application in the context of a web browser

Reporting, Testing and Monitoring

- All these topics are very interconnected.
- We need them all!

- Example – document count for a data source:
  - Report on document growth over time.
  - Test that code changes or new data models will not cause failure to create new document instances
  - Monitor and alert if we reach certain thresholds under normal operations (independents of code changes or deployments).
    - Accidental deletes, accidental duplicates, accidental adding/removing documents from a different refinery

We are gradually adding more sites

- Test setup needs dynamic discovery
  - Cannot expect to remember adding more tests for every site
  - not easy with just point and click systems

- We need something code-friendly

Testing DATA

Cognite is python & API based

- The Cognite applications and GUIs are all built on top of public APIs.
- The project team uses python for all work
  - custom functions
  - Jupyter notebooks
  - custom extractors.

Reflections:

- Write standard python code not tied to a specific external dependency or framework.
- Python built-in “unit tests“ fit well for lots of other types of testing beyond unit tests
- Many other systems have plugins for python unit tests, and it is trivial to add integration to i.e. CDF or to Teams channels.

Example EDMS test

Cognite > test > data > {} edms-sample-row.json > ...

```json
{
    "DWG_Sub_Type_Code": "Loop Diagram",
    "Catalog_Profile_Code": null,
    "Vendor_Document_Number_Code": null,
    "File_Type_Short_Code": null,
    "PSI_Indicator": "N",
    "Modified_Date": "2024-04-09 03:12:00",
    "Geolocation": null,
    "Plant_Name_Short_Code": "SP",
    "Category_Name": null,
```

```python
tables = self.client.raw.tables.list(db_name=self.db)
self.edms_tables =  tables.as_names()
```

```python
def test_expected_columns(self):
    """ Test that all tables have an identical column layout. """
    with open(os.path.join('data', 'edms-sample-row.json'), 'r') as f:
        expected_columns = json.load(f).keys()

        for table in self.edms_tables:
            with self.subTest(table=table):
                rows = self.client.raw.rows.list(self.db, table, limit = 1)
                self.compare_expected_equal_lists(expected_columns, rows[0].columns.keys(), table)
```

Test result [AN & SP]

- V1....

```
----------------------------------------
(table='tbl_indp_edms_files_metadata_AN') SUBFAIL test_edms.py::EDMSTestCase::test_expected_columns - AssertionError: 5 != 0 : tbl_indp_edms_file
s_metadata_AN: These entries were not expected: {'ZDWG AN ADN Sheet:ADN', 'Legacy_Document_ID', 'ZDWG AN ADN:ADN', 'File_Type_Name', 'Pr...
(table='tbl_indp_edms_files_metadata_GB') SUBFAIL test_edms.py::EDMSTestCase::test_expected_columns - AssertionError: 180 != 0 : tbl_indp_edms_fi
les_metadata_GB: These entries were not expected: {'EDM-GBR:Sub-Classification', 'Instrument_SubType', 'EDM-GBR:DCC Classification', 'Ma...
========================================================================== 2 failed, 3 passed, 1 warning in 9.03s ==================================
----------------------------------------
```

- V2 [close!]

```
AssertionError: 2 != 0 : tbl_indp_edms_files_metadata_AN: These entries were not expected: {'ZDWG AN ADN Sheet:ADN', 'ZDWG AN ADN:ADN'}.
 These entries were missing: {'ZDWG SP ADN:ADN', 'ZDWG SP ADN Sheet:ADN'}
```

Example Doclib categories

```python
def test_DocLib(self):
    profile = self.get_profile_result("db_gvlrs253_doclib_gvl", "tbl_gvlrs253_doclib_gvl")
    self.assertGreaterEqual(profile['rowCount'], 1000)

    primary = profile['columns']['primary_class']

    self.assertEqual(primary['nullCount'], 0)

    # Should have a goldilocks number of unique primary classes, not too few, not too many
    self.assertGreater(primary['string']['distinctCount'], 10)
    self.assertLess(primary['string']['distinctCount'], 100)

    # Validate a few cases we know are supposed to be there!
    values = primary['string']['valueCounts'][0]

    self.assertIn('EQUIP FILE', values)
    self.assertIn('ISO', values)
    self.assertIn('INSTRUMENTATION', values)
    self.assertIn('PID', values)
    self.assertIn('ROTATING EQUIPMENT', values)
```

00:00:03

too few, not too many

expanc

```
        self.assertIn('EQUIP FILE', values)
>       self.assertIn('ISO', values)
E       AssertionError: 'ISO' not found in ['INSTRUMENTATION', 'ROTATING EQUIPMENT', 'ELECTRICAL EQUIPMENT',
'MISCELLANEOUS', 'EQUIP FILE', 'SAFETY', 'INST DRAW', '<other>', 'INSPECTION', 'ELEC DRAW', 'FIXED EQUIPMENT']

test_doclib.py:33: AssertionError
```

Html test runner

Summary

39 tests took 00:09:51.

(Un)check the boxes to filter the results.

☑ 22 Failed, ☑ 17 Passed, ☑ 0 Skipped, ☑ 0 Expected failures, ☑

Result	Test
Passed	Cognite/test/test_basics.py::BasicsTestCase::test_connection
Passed	Cognite/test/test_compare_env_counts.py::CompareEnvTestCase::test_DocLib
Passed	Cognite/test/test_compare_env_counts.py::CompareEnvTestCase::test_audit_SI
Passed	Cognite/test/test_compare_env_counts.py::CompareEnvTestCase::test_audit_Salus
Passed	Cognite/test/test_compare_env_counts.py::CompareEnvTestCase::test_iTAR_events
Passed	Cognite/test/test_compare_env_counts.py::CompareEnvTestCase::test_iTAR_worklist
Passed	Cognite/test/test_compare_env_counts.py::CompareEnvTestCase::test_intelex
Passed	Cognite/test/test_doclib.py::DoclibTestCase::test_content_availability
Passed	Cognite/test/test_edms.py::EDMSTestCase::test_content_availability
Passed	Cognite/test/test_edms.py::EDMSTestCase::test_dwg_subtypes
Passed	Cognite/test/test_edms.py::EDMSTestCase::test_expected_columns
Passed	Cognite/test/test_object_counts.py::ObjectCountTestCase::test_audit_SI
Passed	Cognite/test/test_object_counts.py::ObjectCountTestCase::test_audit_Salus
Passed	Cognite/test/test_object_counts.py::ObjectCountTestCase::test_iTAR_events
Passed	Cognite/test/test_object_counts.py::ObjectCountTestCase::test_iTAR_worklist
Passed	Cognite/test/test_object_counts.py::ObjectCountTestCase::test_intelex

```
>       self.assertIn('tbl_gvlrs253_doclib_gvl', names)
E       AssertionError: 'tbl_gvlrs253_doclib_gvl' not found in ['annotation_documents_tags', 'files_state_store',
'annotation_documents_docs', 'tbl_gvlrs253_doclib_gvl_test_6', 'eric_dl_test_dump',
'tbl_gvlrs253_doclib_gvl_test_10', 'iso_summary', 'tbl_gvlrs253_doclib_gvl_test',
'tbl_states_gvlrs253_doclib_gvl', 'tbl_gvlrs253_doclib_gvl_test_5']

test_basics.py:36: AssertionError
```

Testing the full App end2end

Marathon runs in a secure environment

- The Marathon environment is locked down with private link.
- Getting permissions right for all use cases have proven to be a bit challenging.

 Don’t even consider lots of easy web based solutions

Marathon is ~100% Microsoft

- We need something Microsoft friendly!

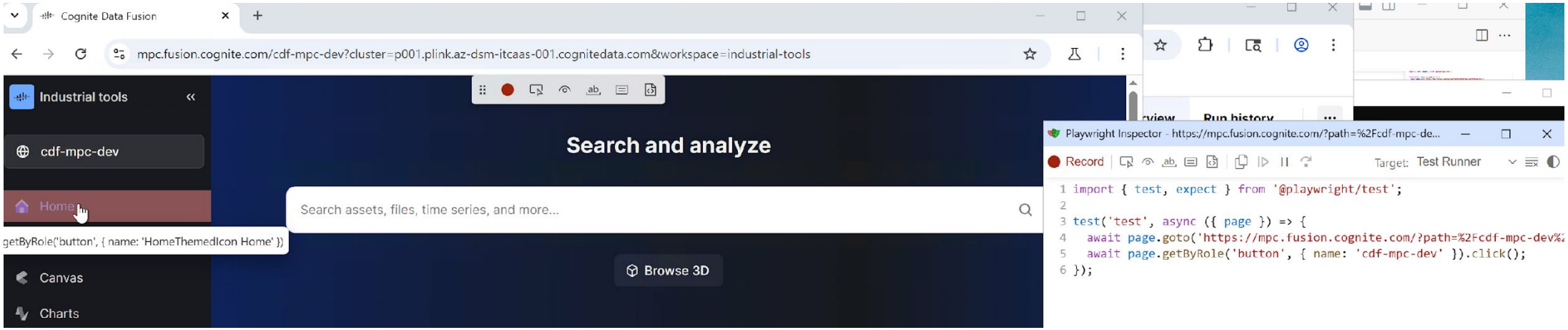
Playwright

- https://playwright.dev/

- Open Source by Microsoft.
- Supports “easy browser coding”
- Attempts to overcome limitations of tools like Puppeteer and Selenium.

Test creation
It produces code as you click in GUI

```
C:\Users\M4C5\Repos\test>npx playwright codegen
```



```
3
4  test('test', async ({ page }) => {
5    await page.goto('https://mpc.fusion.cognite.com/?path=%2Fcdf-mpc-dev%2F&search='); — 1792ms
6    await page.getByRole('button', { name: 'cdf-mpc-dev' }).click(); — 24902ms
7    await page.getByRole('button', { name: 'ChartsSubAppThemedIcon Charts' }).click(); — 3953ms
8    await page.getByTestId('public-charts-tab').click(); — 4896ms
9    await page.getByRole('button', { name: 'Tri_County_03' }).click(); — 307ms
10   await page.getByRole('button', { name: 'Timeseries EyeShowIcon Tri_County_Station.Tri_County_03.3rd Stage Discharge' }).click();
11
12   const tr = page.locator('tr:has-text("Tri_County_Station.Tri_County_03.3rd Stage Discharge Press")');
13   expect(tr).not.toBeUndefined(); — 1ms
14
15   expect(await tr.count()).toEqual(1); — 1ms (ran 2x)
16
17   const td = tr.locator('td:nth-child(6)');
18
19   const text = await td.textContent(); — 9ms
20
21   expect(text).not.toBeNull(); — 0ms
22
23   expect(parseInt(text!)).toBeGreaterThan(20); — 1ms
24
25 });
```

mpc.fusion.cognite.com/cdf-mpc-dev/chart

Cognite Data Fusion

Industrial tools

cdf-mpc-dev

- Home
- Search
- Canvas
- Charts
- Custom apps BETA
- Agent library BETA
- Help and resources
- Borg, Ragnar

Charts / Tri_County_

View only. Duplicate to

Auto-fit axes

STYLE	NAME	STATUS	TAG	DESCRIPTION	MIN	MAX	MEAN	UNIT
	Tri_County_Station.Tri_County_03.3rd Stage Discharge Press		C-270		114	114	114	psig *
	Tri_County_Station.Tri_County_03.3rd Stage Suction Press		C-270		0	0	0	psig *
	Tri_County_Station.Tri_County_03.2nd Stage Suction Press		C-270		0	0	0	psig *
	Tri_County_Station.Tri_County_03.2nd Stage Discharge Press		C-270		107	107	107	psig *